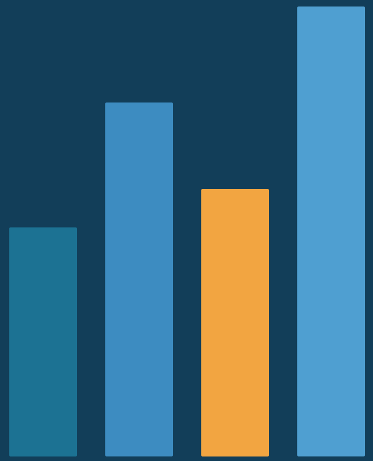
IoT-Enabled Supply Chain Monitoring

Cold-chain risk detection · environmental visibility · supply-chain transparency

10+ analytical questions

Plotly-only notebook

Streamlit dashboard



Abubakar Umar Dangi
MSc Data Science

Project alignment

The prototype supports an analytical supply-chain story, but provenance must be transparent.

Brief requirements

Real-world, rich, varied dataset; 10+ analytical questions; Plotly-only explanatory visuals; curated interactive Streamlit dashboard; final presentation exported to PDF.

Current asset

The existing dashboard is a strong Streamlit proof-of-concept: route map, threshold alerts, cumulative CO2, alert log, and detection-latency comparison.

Integrity decision

The uploaded sensor records are labelled simulated_prototype. Final submission should add a verified real-world dataset and keep the prototype as a scenario layer.

580
sensor readings

4
shipments

3.86 h
avg latency saved

100%
runs with alerts

Data pipeline

A clean separation between raw prototype files, processed analysis tables, notebook, dashboard, and deck.

Raw uploads

shipment CSVs
batch_results.json
original dashboard

→

Prepare data

clean table
summary table
latency experiment

→

Notebook

EDA
10 analytical questions
Plotly figures

→

Dashboard

curated tabs
filters
interactive visuals

→

Presentation

insights
conclusions
dashboard screenshots

Processed tables created

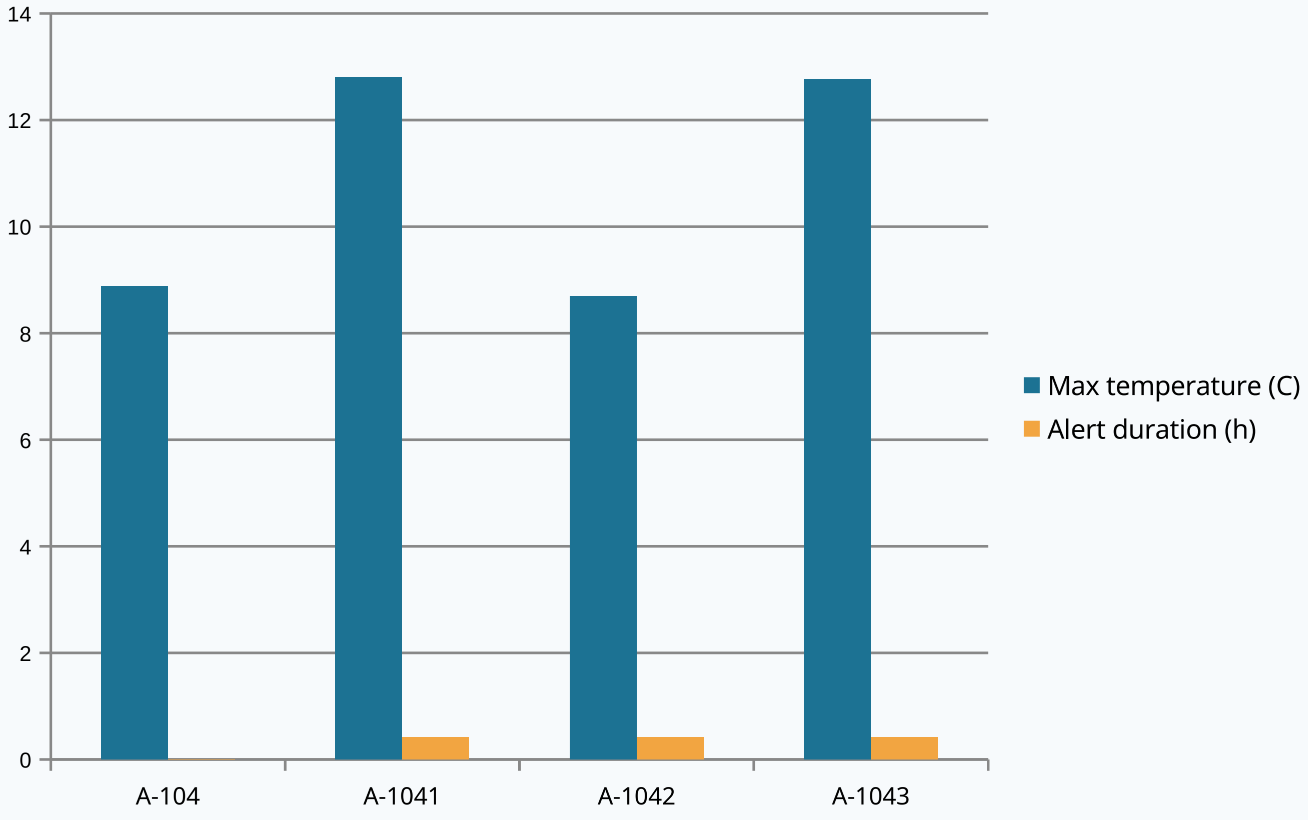
clean_supply_chain_prototype.csv · shipment_summary_prototype.csv · batch_results.csv

Next improvement

Add a real-world dataset such as UCI Daily Demand Forecasting Orders or a verified cold-chain sensor dataset, then connect it to the same dashboard architecture.

Sensor risk by shipment

Maximum temperature and unsafe exposure duration identify which shipments need attention.



Takeaway

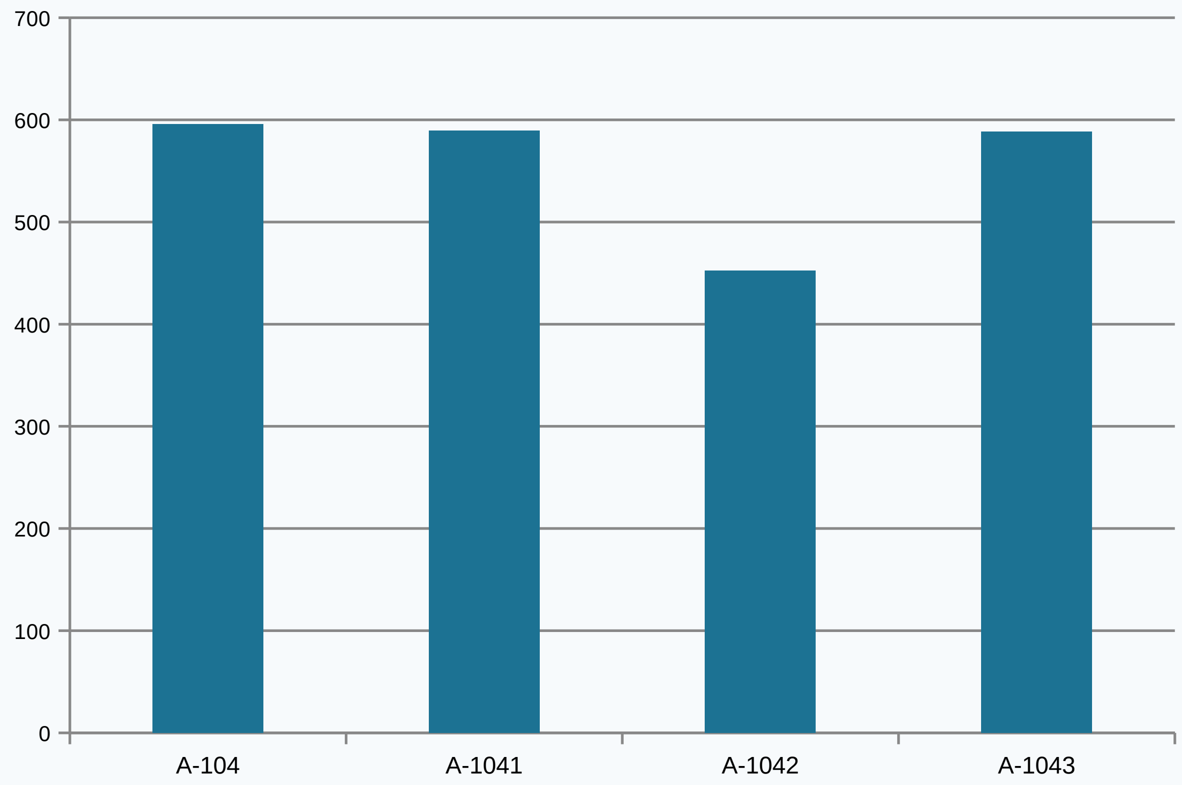
Risk is not just whether an alert happens. The key is how intense the excursion becomes and how long the shipment stays outside its threshold.

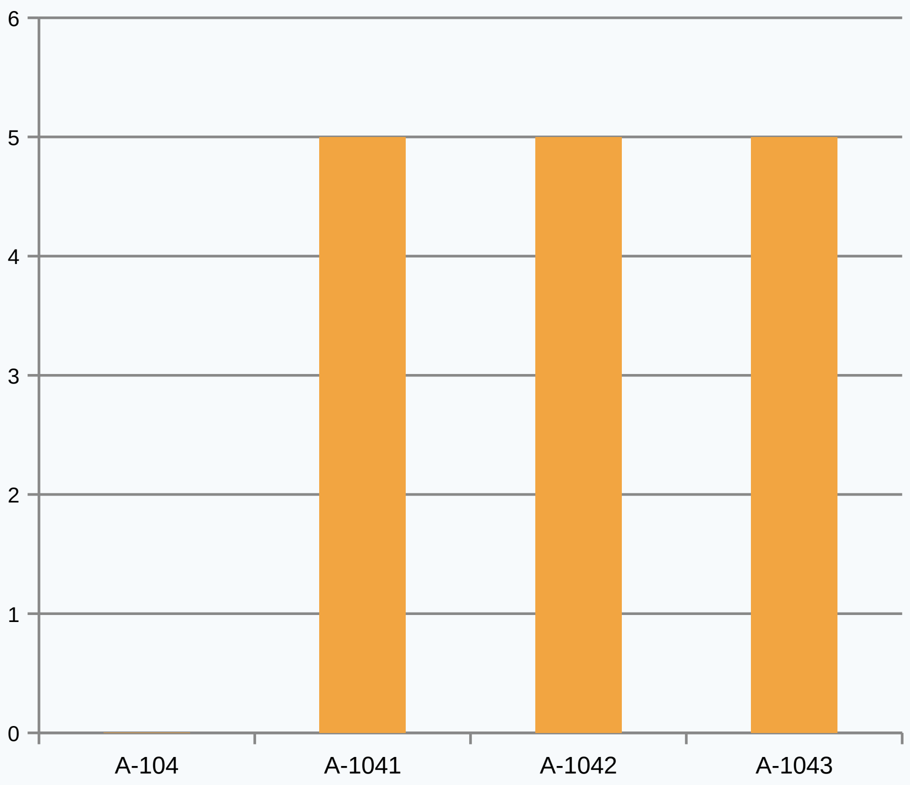
Managerial use

Rank shipments for inspection and intervention using maximum temperature, alert duration, and emissions rather than a single alert flag.

Emissions visibility

The prototype links operational movement to environmental impact.

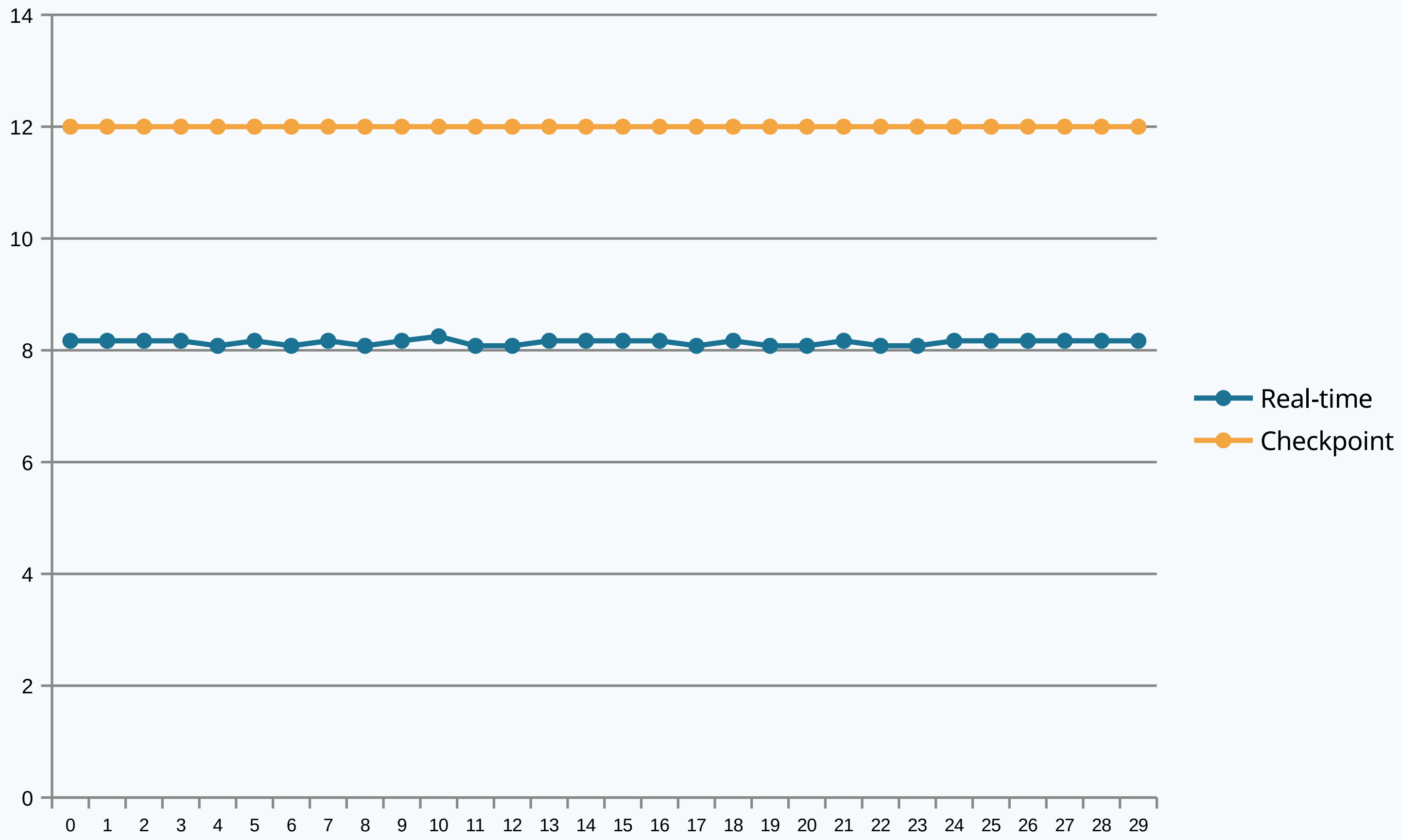




Two operational lenses belong together: climate impact and cold-chain safety. This makes the dashboard useful for both sustainability and quality control.

Real-time detection advantage

Across repeated runs, real-time monitoring detects threshold breaches around 3.86 hours before checkpoint inspection.



3.86 h

mean hours saved

100%

alert occurrence rate

Interpretation

Earlier visibility creates time for corrective action: rerouting, contacting the carrier, checking refrigeration equipment, or prioritizing receiving inspection.

Dashboard design

A curated Streamlit app, not a dump of every figure.

Executive Overview	KPIs, headline trends, summary table
Journey & Sensors	Route map, CO2 accumulation, interval emissions
Alerts & Risk	Threshold breaches by stage and shipment
Detection Speed	Real-time vs checkpoint comparison
Data Provenance	Clear disclosure and next dataset step

Primary filters: shipment ID · threshold · alert-only toggle

Final project package

Everything needed to continue the submission workflow has been assembled.

Ready now

Cleaned prototype datasets, data-preparation script, executable analysis notebook, Streamlit dashboard, README, presentation deck, and zipped project package.

Before final submission

Add the real-world dataset, re-run the notebook, insert dashboard screenshots, deploy to Streamlit Community Cloud, then export the deck to PDF.

Notebook

Dashboard

Deck

GitHub

Streamlit

PDF submit

One dataset story · ten analytical questions · one interactive dashboard